

1 Linear Combinations

$$S * Q + Q * S$$

$$+ \begin{pmatrix} \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+2 \cdot |aaaa0000| + 2 \cdot |aa0000aa| + 2 \cdot |a0a00a0a| + 2 \cdot |a00a0aa0| + 2 \cdot |0aa0a00a| + 2 \cdot |0a0aa0a0|$$

$$+2 \cdot |00aaaa00| + 2 \cdot |0000aaaa|$$

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right|$$

$$+2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|$$

$$S * Q - Q * S$$

$$+ \begin{pmatrix} \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$-2 \cdot |aaa0000a| - 2 \cdot |aa0a00a0| - 2 \cdot |a0aa0a00| - 2 \cdot |a0000aaa| - 2 \cdot |0aaaa000| - 2 \cdot |0a00a0aa|$$

$$-2 \cdot |00a0aa0a| - 2 \cdot |000aaaa0|$$

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right|$$

$$-2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+ |a2a20202| - |a20202a2| - |02a2a202| + |0202a2a2| - 2 \cdot |aaaa0000| - 2 \cdot |a0a00a0a| \\ + 2 \cdot |aa0000aa| + 2 \cdot |a00a0aa0| + 2 \cdot |0aa0a00a| + 2 \cdot |00aaaa00| - 2 \cdot |0a0aa0a0| - 2 \cdot |0000aaaa| \\ + |2a2a2020| - |2a20202a| - |202a2a20| + |20202a2a|$$

or, expressed in terms of symmetry:

$$+ \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right|$$

$$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+ 2 \cdot |a2a20202| - 2 \cdot |a20202a2| - 2 \cdot |02a2a202| + 2 \cdot |0202a2a2| - 2 \cdot |2a2a2020| + 2 \cdot |2a20202a| \\ + 2 \cdot |202a2a20| - 2 \cdot |20202a2a|$$

or, expressed in terms of symmetry:

$$+ 2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| \\ + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right|$$

$$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$-4 \cdot |aaa0000a| - 4 \cdot |a0aa0a00| + 4 \cdot |aa0a00a0| + 4 \cdot |a0000aaa| + 4 \cdot |0aaaa000| + 4 \cdot |00a0aa0a|$$

$$-4 \cdot |0a00a0aa| - 4 \cdot |000aaaa0|$$

or, expressed in terms of symmetry:

$$-4 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\ - 4 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

$$+2 \cdot |aaa20002| - 2 \cdot |a0020aa2| - 2 \cdot |0aa2a002| + 2 \cdot |0002aaa2| - 2 \cdot |a2aa0200| + 2 \cdot |a20a02a0|$$

$$+2 \cdot |02a0a20a| - 2 \cdot |0200a2aa| - 2 \cdot |aa2a0020| + 2 \cdot |0a20a02a| + 2 \cdot |a02a0a20| - 2 \cdot |0020aa2a|$$

$$+2 \cdot |2aaa2000| - 2 \cdot |2aa0200a| - 2 \cdot |200a2aa0| + 2 \cdot |20002aaa|$$

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\ +2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\ +2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|$$

$$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

$$-2 \cdot |a0a20a02| + 2 \cdot |aa0200a2| + 2 \cdot |00a2aa02| - 2 \cdot |0a02a0a2| + 2 \cdot |a2a0020a| - 2 \cdot |a20002aa|$$

$$-2 \cdot |02aaa200| + 2 \cdot |020aa2a0| + 2 \cdot |0a2aa020| - 2 \cdot |aa20002a| - 2 \cdot |002aaa20| + 2 \cdot |a0200a2a|$$

$$-2 \cdot |2a0a20a0| + 2 \cdot |2a0020aa| + 2 \cdot |20aa2a00| - 2 \cdot |20a02a0a|$$

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\ -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\ -2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|$$

$$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

$$+2 \cdot |aaa20002| - 2 \cdot |a0020aa2| - 2 \cdot |0aa2a002| + 2 \cdot |0002aaa2| + 2 \cdot |a2aa0200| - 2 \cdot |a20a02a0|$$

$$-2 \cdot |02a0a20a| + 2 \cdot |0200a2aa| - 2 \cdot |aa2a0020| + 2 \cdot |0a20a02a| + 2 \cdot |a02a0a20| - 2 \cdot |0020aa2a|$$

$$-2 \cdot |2aaa2000| + 2 \cdot |2aa0200a| + 2 \cdot |200a2aa0| - 2 \cdot |20002aaa|$$

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|$$

$$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

$$-2 \cdot |a0a20a02| + 2 \cdot |aa0200a2| + 2 \cdot |00a2aa02| - 2 \cdot |0a02a0a2| - 2 \cdot |a2a0020a| + 2 \cdot |a20002aa|$$

$$+2 \cdot |02aaa200| - 2 \cdot |020aa2a0| + 2 \cdot |0a2aa020| - 2 \cdot |aa20002a| - 2 \cdot |002aaa20| + 2 \cdot |a0200a2a|$$

$$+2 \cdot |2a0a20a0| - 2 \cdot |2a0020aa| - 2 \cdot |20aa2a00| + 2 \cdot |20a02a0a|$$

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\ +2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\ +2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|$$

$$e^3 b_{2u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ -\uparrow & \uparrow\uparrow & \uparrow\uparrow & -\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ -\uparrow & \uparrow\uparrow & -\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+ |a2a20202| - |a20202a2| - |02a2a202| + |0202a2a2| + 2 \cdot |aaaa0000| + 2 \cdot |a0a00a0a|$$

$$- 2 \cdot |aa0000aa| - 2 \cdot |a00a0aa0| - 2 \cdot |0aa0a00a| - 2 \cdot |00aaaa00| + 2 \cdot |0a0aa0a0| + 2 \cdot |0000aaaa|$$

$$+ |2a2a2020| - |2a20202a| - |202a2a20| + |20202a2a|$$

or, expressed in terms of symmetry:

$$\begin{aligned} & + \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ & - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\ & + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right| \end{aligned}$$

$$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

$$+2 \cdot |aaa20002| - 2 \cdot |a0020aa2| - 2 \cdot |0aa2a002| + 2 \cdot |0002aaa2| - 2 \cdot |a2aa0200| + 2 \cdot |a20a02a0|$$

$$+2 \cdot |02a0a20a| - 2 \cdot |0200a2aa| + 2 \cdot |aa2a0020| - 2 \cdot |0a20a02a| - 2 \cdot |a02a0a20| + 2 \cdot |0020aa2a|$$

$$-2 \cdot |2aaa2000| + 2 \cdot |2aa0200a| + 2 \cdot |200a2aa0| - 2 \cdot |20002aaa|$$

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\ +2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|$$

$$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

$$-2 \cdot |a0a20a02| + 2 \cdot |aa0200a2| + 2 \cdot |00a2aa02| - 2 \cdot |0a02a0a2| + 2 \cdot |a2a0020a| - 2 \cdot |a20002aa|$$

$$-2 \cdot |02aaa200| + 2 \cdot |020aa2a0| - 2 \cdot |0a2aa020| + 2 \cdot |aa20002a| + 2 \cdot |002aaa20| - 2 \cdot |a0200a2a|$$

$$+ 2 \cdot |2a0a20a0| - 2 \cdot |2a0020aa| - 2 \cdot |20aa2a00| + 2 \cdot |20a02a0a|$$

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\ -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|$$

$$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+2 \cdot |aaa20002| - 2 \cdot |a0020aa2| - 2 \cdot |0aa2a002| + 2 \cdot |0002aaa2| + 2 \cdot |a2aa0200| - 2 \cdot |a20a02a0|$$

$$-2 \cdot |02a0a20a| + 2 \cdot |0200a2aa| + 2 \cdot |aa2a0020| - 2 \cdot |0a20a02a| - 2 \cdot |a02a0a20| + 2 \cdot |0020aa2a|$$

$$+2 \cdot |2aaa2000| - 2 \cdot |2aa0200a| - 2 \cdot |200a2aa0| + 2 \cdot |20002aaa|$$

or, expressed in terms of symmetry:

$$+2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\ +2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|$$

$$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI vectors:

$$-2 \cdot |a0a20a02| + 2 \cdot |aa0200a2| + 2 \cdot |00a2aa02| - 2 \cdot |0a02a0a2| - 2 \cdot |a2a0020a| + 2 \cdot |a20002aa|$$

$$+ 2 \cdot |02aaa200| - 2 \cdot |020aa2a0| - 2 \cdot |0a2aa020| + 2 \cdot |aa20002a| + 2 \cdot |002aaa20| - 2 \cdot |a0200a2a|$$

$$- 2 \cdot |2a0a20a0| + 2 \cdot |2a0020aa| + 2 \cdot |20aa2a00| - 2 \cdot |20a02a0a|$$

or, expressed in terms of symmetry:

$$-2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\ - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|$$

$$e^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

written in CI vectors:

$$+ |aa220022| - |a0220a22| - |0a22a022| + |0022aa22| - 2 \cdot |aaaa0000| - 2 \cdot |aa0000aa|$$

$$+ 2 \cdot |a0a00a0a| + 2 \cdot |a00a0aa0| + 2 \cdot |0aa0a00a| + 2 \cdot |0a0aa0a0| - 2 \cdot |00aaaa00| - 2 \cdot |0000aaaa|$$

$$+ |22aa2200| - |22a0220a| - |220a22a0| + |220022aa|$$

or, expressed in terms of symmetry:

$$\begin{aligned} & + \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\ & + \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right| \end{aligned}$$

$$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$$

$$\begin{aligned} & + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} \\ & - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \end{aligned}$$

written in CI vectors:

$$+ 2 \cdot |aa220022| - 2 \cdot |a0220a22| - 2 \cdot |0a22a022| + 2 \cdot |0022aa22| - 2 \cdot |22aa2200| + 2 \cdot |22a0220a|$$

$$+ 2 \cdot |220a22a0| - 2 \cdot |220022aa|$$

or, expressed in terms of symmetry:

$$\begin{aligned} & + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| \\ & + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right| \end{aligned}$$

$$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

written in CI vectors:

$$-4 \cdot |aaa0000a| - 4 \cdot |aa0a00a0| + 4 \cdot |a0aa0a00| + 4 \cdot |a0000aaa| + 4 \cdot |0aaaa000| + 4 \cdot |0a00a0aa|$$

$$-4 \cdot |00a0aa0a| - 4 \cdot |000aaaa0|$$

or, expressed in terms of symmetry:

$$-4 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ - 4 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|$$

$$i^3 b_{3u} * i^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

written in CI vectors:

$$+ |aa220022| - |a0220a22| - |0a22a022| + |0022aa22| + 2 \cdot |aaaa0000| + 2 \cdot |aa0000aa|$$

$$-2 \cdot |a0a00a0a| - 2 \cdot |a00a0aa0| - 2 \cdot |0aa0a00a| - 2 \cdot |0a0aa0a0| + 2 \cdot |00aaaa00| + 2 \cdot |0000aaaa|$$

$$+ |22aa2200| - |22a0220a| - |220a22a0| + |220022aa|$$

or, expressed in terms of symmetry:

$$+ \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\ + \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right|$$
